Math 3
VII-R Solve Equations Algebraically: Review

Name:_____
Date: _____ Per: ___

For problems 1 - 8, solve for x.

1) $3x - 4 = 2x + 8 - 5x$

2) $2 - 15x = 5(3x + 2)$

3) $x - 4 = \frac{2}{3}x - 8 + \frac{1}{6}x$

4) $8x = 4(2x + 1)$

5) $6.2x + 5 - 1.4x = 4.8x + 6$

6) $5.4x + 14.6 - 10.1x = 12.8x - 3.5x - 0.6$

7) $14 - \frac{1}{4}(x - 10) = \frac{5}{6}(x + 15)$

8) $2(3x + 2) = \frac{1}{2}(12x + 8)$

Turn Over

9) The membership fee for joining a camping group is $45. A campsite charges $35 per night for members of this camping group and nonmembers $40 per night. After how many nights would a nonmember and a member pay the same amount?

10) In the 2020 NFC Championship Game, the 49ers scored 1 less than ⅔ the total points scored in the game. The Packers scored 56 points less than 4/3 the total points scored in the game.

Write and solve an equation to determine the total points scored in the game.

11) Who won the game in item 10? What was the final score?

For problems 12 - 15, simplify the expression.

12) $(x^9)^2 \bullet x^{14}$

13) $\frac{(2x^3y^{-10})^4}{4x^3y^{-3}}$

14) $(6xy^9)(4xyz^0)$

15) $\frac{3}{2x^{-9}}$

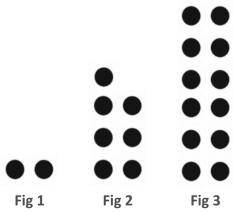
16) If you double 2^{200} what is the result? Explain.

For problems 17 and 18, simplify the expression. Write your answer in scientific notation.

17) $5.7 \times 10^{100} - 4 \times 10^{98}$

18) $(5.7 \times 10^{100})(4 \times 10^{98})$

For problems 19 and 20.
Use the figures to the right.



19) If n = figure number, write an expression to determine the number of dots in the nth figure.

20) How many dots are in the 155th figure?